

The following content is based on

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§3.1 Submersion of complex manifold

The background of this section is actually the context of family local index theorem of Bismut. The necessary geometric data are listed as below.

1. Two complex manifolds M and B . A proper holomorphic submersion $\pi : M \rightarrow B$ with compact fiber Z , i.e. locally we have a smooth diffeomorphism

$$\pi^{-1}(U) \simeq U \times Z$$

2. A metric g^{TZ} on TZ , the vertical tangent subbundle of TM
3. A horizontal subbundle of TM such that $TM = T^H M \oplus TZ$.

Clearly, $T^H M \xrightarrow{\pi_*} TB$ is a bijection. It induces an isomorphism $T^H M \simeq \pi^*(TB)$. Let J^{TZ} be a complex structure on TZ , g^{TZ} be a J^{TZ} -invariant Riemannian metric on TZ , i.e.

$$\forall u, v \in C^\infty(M, TZ), g^{TZ}(u, v) = g^{TZ}(J^{TZ}u, J^{TZ}v)$$

Extend $\mathbb{C}$ -linearly g^{TZ} , denoted also by g^{TZ} . Let J be a complex structure on TM which is compatible with J^{TZ} , that is $J|_{TZ} = J^{TZ}$. Consequently, J preserve TZ . We work under the assumption that the horizontal subbundle $T^H M$ is also J -invariant.

Definition. The triple $(\pi, g^{TZ}, T^H M)$ is a Kahler fibration if there exists a real closed $(1, 1)$ -form $\omega \in \Omega^{1,1}(M)$ ("real" means $\bar{\omega} = \omega$, closed means $d\omega = 0$) such that

Condition (1) $T^H M \perp TZ$ w.r.t ω , i.e.

$$\forall U \in C^\infty(M, T^H M), Y \in C^\infty(M, TZ), \omega(U, Y) = 0$$

Condition (2) $\forall X, Y \in C^\infty(M, TZ), \omega(X, Y) = g^{TZ}(X, JY)$

If a real closed $(1, 1)$ -form satisfies the conditions above, we say that it is associated with the Kahler foliation $(\pi, g^{TZ}, T^H M)$. Conversely, if we have such a form, then the horizontal subbundle can be reconstructed by

$$T^H M = \bigcup_{x \in M} \{Y \in T_x M | \omega(X, Y) = 0 \ \forall X \in T_x Z\}.$$

It is a C^∞ subbundle of TM such that $TM = T^H M \oplus TZ$.

Proposition. For a given Kahler fibration $(\pi, g^{TZ}, T^H M)$ with an associated form ω , another form ω' is also associated with $(\pi, g^{TZ}, T^H M)$ (defines the same geometric data) if and only if there exists a closed form $\eta \in \Omega^{1,1}(B)$ such that $\omega - \omega' = \pi^* \eta$.

Proof. Set $\sigma = \omega - \omega'$. By Condition (1), $\forall U \in C^\infty(M, T^H M)$, $Y \in C^\infty(M, TZ)$, $\sigma(U, Y) = 0$. For any $X, Y \in C^\infty(M, TZ)$,

$$\sigma(X, Y) = (\omega - \omega')(X, Y) = g^{TZ}(X, JY) - g^{TZ}(X, JY) = 0.$$

Thus, $\forall Y \in C^\infty(M, TZ)$, $i_Y \sigma = 0$.

$$\begin{array}{c} \mathcal{L}_Y \sigma \xrightarrow[\text{Formula}]{\text{Cartan's}} i_Y d\sigma + di_Y \sigma \\ \xrightarrow[\text{Formula}]{d\omega = d\omega' = 0} di_Y \sigma = 0. \end{array}$$

□

Example. Let M, B be two complex manifolds and $\pi : M \rightarrow B$ be a proper holomorphic submersion with compact fiber Z . Denote by J the complex structure on M and g^{TM} the Hermitian metric obtained by $\mathbb{C}$ -bilinear extension of a Riemannian metric on $T_{\mathbb{R}} M$. Our form associated to (M, J, g^{TM}) is given by $\omega(\cdot, \cdot) = g^{TM}((\cdot), J(\cdot))$. Attention here J locates at the second component. While the standard fundamental form's J locates at the first component. In addition, the complex structure $J|_{TZ}$ of J restricted on TZ is a complex structure on TZ . In other words, J has a block matrix form

$$J = \begin{bmatrix} J^{TZ} & 0 \\ 0 & * \end{bmatrix} TZ$$

Thus, J preserve TZ . Assume that ω is a Kahler form, i.e. $d\omega = (\partial + \bar{\partial})\omega = 0$. Restrict g^{TM} on the subbundle TZ , we have a $\mathbb{C}$ -linearly extended Riemannian metric g^{TZ} . The Riemannian metric g^{TZ} can also be defined from the Kahler form ω by $g^{TZ}(\cdot, \cdot) = \omega(J\cdot, (\cdot))$. We show that g^{TZ} is J -invariant. Indeed, $\forall X, Y \in C^\infty(M, TZ)$,

$$g^{TZ}(JX, JY) = \omega(J^2 X, JY) = -\omega(X, JY) \xrightarrow[g^{TM} \text{ is } J \text{ invariant}]{\omega(JX, Y) + \omega(X, JY) = 0} \omega(JX, Y) = g^{TZ}(X, Y)$$

Define $T^H M = (TZ)^\perp$ with respect to the $\mathbb{C}$ -linearly extended Riemannian metric g^{TM} . We show that $T^H M$ is J -invariant. Indeed, $\forall X \in C^\infty(M, T^H M), Y \in C^\infty(M, TZ)$,

$$g^{TM}(JU, Y) = g^{TM}(J^2 U, JY) = -g^{TM}(U, JY) \xrightarrow[J \text{ invariant}]{TZ \text{ is}} 0.$$

Remark. Locally on an open set U of B , there is a diffeomorphism $\pi^{-1}(U) \simeq U \times Z$. Take a Kahler form η^U on U . We know that $\eta(J\cdot, \cdot)$ is a $\mathbb{C}$ -linearly extended Riemannian metric. In local coordinates $(z_1, \dots, z_l)$ of Z and $(t_1, \dots, t_m)$ of U , the following matrix

$$\begin{bmatrix} \sqrt{-1}\omega(\frac{\partial}{\partial z_i}, \frac{\partial}{\partial \bar{z}_j}) & \vdots & 0 \\ \hline 0 & \vdots & \sqrt{-1}\omega(\frac{\partial}{\partial t_k}, \frac{\partial}{\partial t_l}) \end{bmatrix}$$

is Hermitian and $\begin{bmatrix} \sqrt{-1}\omega(\frac{\partial}{\partial z_i}, \frac{\partial}{\partial \bar{z}_i}) \end{bmatrix}$ is positive definite. For $\lambda > 0$, let $\omega_\lambda = \omega + \lambda\pi^*\eta^U$, the corresponding Hermitian matrix is

$$\begin{bmatrix} \sqrt{-1}\omega(\frac{\partial}{\partial z_i}, \frac{\partial}{\partial \bar{z}_j}) & \vdots & 0 \\ \hline 0 & \vdots & \sqrt{-1}\omega(\frac{\partial}{\partial t_k}, \frac{\partial}{\partial t_l}) + \lambda\sqrt{-1}\eta^U(\frac{\partial}{\partial t_k}, \frac{\partial}{\partial t_l}) \end{bmatrix}.$$

When λ is large enough, this Hermitian metric is positive definite, thus defines locally a Riemannian metric. Since Z is compact, we can take an uniform λ large enough such that ω_λ defines a Riemannian metric (denote simply by g) on $\pi^{-1}(U)$ whose fundamental form (our J is at second component) is exactly ω_λ . To prove the metric g is J -invariant, which is equivalent to

$$\forall X, Y \in C^\infty(\pi^{-1}U, T\pi^{-1}U), \quad \omega_\lambda(JX, Y) + \omega(X, JY) = 0$$

As J preserve TZ and also preserve $T^H M$ (Assumption at the beginning), it only need to consider X, Y locate simultaneously in $C^\infty(\pi^{-1}(U), TZ|_{\pi^{-1}(U)})$ or in $C^\infty(\pi^{-1}(U), T^H M|_{\pi^{-1}(U)})$. By the geometric setting (Condition 2) and the fact that $\pi^*\eta^U$ is the pullback of a Kahler form, we know g is J -invariant. Furthermore, ω_λ is compatible with the initial geometrical setting, thus by replacing ω by ω_λ we can assume that $\pi : M \rightarrow B$ is locally Kahler.

Let $(\pi, g^{TZ}, T^H M)$ be a Kahler fibration. For a vector field $U \in C^\infty(B, TB)$ on the base manifold, set $U^H \in C^\infty(M, T^H M)$ the horizontal lift under the isomorphism $T^H M \xrightarrow{\pi_*} TB$. Let

$$P^{TZ} : TM = T^H M \oplus TZ \rightarrow TZ$$

denote the projection. Recall that fiberwise holomorphic tangent bundle $T^{1,0}Z$ has the Chern connection $\nabla^{T^{1,0}Z}$ with respect to the Hermitian metric $h^{T^{1,0}Z}$ induced by g^{TZ} . For $\bar{U} \in C^\infty(M, T^{0,1}Z)$, $X \in C^\infty(M, TM)$, define $\nabla_X^{T^{0,1}Z}\bar{U} = \overline{\nabla_X^{T^{1,0}Z}U}$. Set $\nabla^{TZ} = \nabla^{T^{1,0}Z} \oplus \nabla^{T^{0,1}Z}$, it is a Hermitian connection on TZ .

As (Z, g^{TZ}) is a Kahler manifold with Kahler form $\omega|_{TZ \times TZ}$, we know that ∇^{TZ} is just the Levi-Civita connection on TZ along Z (along Z means derive along the direction of Z).

Definition. For $U, V \in C^\infty(B, TB)$, $X \in C^\infty(M, TZ)$, define $T : TM \times TM \rightarrow TZ$ such that $T(X, \cdot) = 0$ and

$$\begin{aligned} T(U^H, V^H) &= -P^{TZ}[U^H, V^H] \\ T(U^H, X) &= \nabla_{U^H}^{TZ} X - [U^H, X] \end{aligned}$$

Check that $[U^H, X] \in C^\infty(M, TZ)$. Indeed, for any $f \in C^\infty(B)$,

$$[U^H, X](\pi^* f) = U^H \underbrace{X(\pi^* f)}_{=0} - XU^H(\pi^* f) = -X\pi^*(Uf) = 0.$$

Thus, the vector field $[U^H, X]$ is vertical and T is well-defined.

Proposition. $T(\cdot, \cdot)$ is $(1, 1)$ -type real tensor.

Proof. First, we check that T is a real form.

$$T(\overline{U^H}, \overline{V^H}) = -P^{TZ}[\overline{U^H}, \overline{V^H}] = -P^{TZ}[\overline{U^H}, \overline{V^H}].$$

As

$$\nabla_{\overline{U^H}}^{TZ} \overline{X} = \nabla_{\overline{U^H}}^{T^{1,0}Z} \overline{X}^{(1,0)} + \overline{\nabla_{U^H}^{T^{0,1}Z} X^{(0,1)}} = \overline{\nabla_{U^H}^{T^{1,0}Z} X^{(1,0)}} + \nabla_{\overline{U^H}}^{T^{0,1}Z} \overline{X}^{(0,1)} = \overline{\nabla_{U^H}^{T^{1,0}Z} X^{(1,0)}} + \overline{\nabla_{U^H}^{T^{0,1}Z} X^{(0,1)}} = \overline{\nabla_{U^H}^{TZ} X},$$

we have $T(\overline{U^H}, \overline{X}) = \overline{T(U^H, X)}$. Therefore, T is real. As the proposition is local, we may work on $\pi^{-1}(U)$ where U is a local chart U of B . By the remark above, we can see M as $\pi^{-1}(U)$ and assume M to be Kahler w.r.t. some Riemannian metric g^{TM} . Similarly, we have the Chern connection $\nabla^{T^{1,0}M}$ on the holomorphic vector bundle $T^{1,0}M$. We prove that $\nabla^{T^{1,0}Z} = P^{TZ} \nabla^{T^{1,0}M}$. As $\nabla^{T^{1,0}M}$ is Hermitian, for any $U \in C^\infty(M, T^{1,0}M)$, $A, B \in C^\infty(M, T^{1,0}M)$,

$$U \langle A, B \rangle_{T^{1,0}M} = \langle \nabla_U^{T^{1,0}M} A, B \rangle_{T^{1,0}M} + \langle A, \nabla_{\overline{U}}^{T^{1,0}M} B \rangle_{T^{1,0}M}.$$

As $T_{\mathbb{R}}M \otimes \mathbb{C} = T^{1,0}M \oplus T^{0,1}M$, $T_{\mathbb{R}}Z \otimes \mathbb{C} = T^{1,0}Z \oplus T^{0,1}Z$, P^{TZ} projects $T^{1,0}M$ to $T^{1,0}Z$. By the construction of the Kahler form in Remark, we know that TZ and $T^H M$ are orthogonal w.r.t the metric g given by $(\omega + \lambda \pi^* \eta^U)$. Restrict the metric g to TZ can obtain g^{TZ} , so restrict induced Hermitian metric $\langle \cdot, \cdot \rangle_{T^{1,0}M}$ to $T^{1,0}Z$ can exactly obtain the Hermitian metric $\langle \cdot, \cdot \rangle_{T^{1,0}Z}$. Therefore, in the equation above, restricting $A, B \in C^\infty(M, T^{1,0}Z)$, we get

$$U \langle A, B \rangle_{T^{1,0}Z} = \langle P^{TZ} \nabla_U^{T^{1,0}M} A, B \rangle_{T^{1,0}Z} + \langle A, \nabla_{\overline{U}}^{T^{1,0}M} B \rangle_{T^{1,0}Z}.$$

Thus, $P^{TZ} \nabla^{T^{1,0}M}$ is a Hermitian connection on $T^{1,0}Z$. And

$$\left(P^{TZ} \nabla^{T^{1,0}M} \right)^{(0,1)} = P^{TZ} \left(\nabla^{T^{1,0}M} \right)^{(0,1)} = P^{TZ} \overline{\partial}^{T^{1,0}M} \frac{T^{1,0}M = T^{H,(1,0)}M \oplus T^{1,0}Z}{P^{TZ}: T^{1,0}M \rightarrow T^{1,0}Z} \overline{\partial}^{T^{1,0}Z}$$

tells us that $P^{TZ} \nabla^{T^{1,0}M}$ is holomorphic, so $P^{TZ} \nabla^{T^{1,0}M}$ is the Chern connection. By uniqueness of the Chern connection, $\nabla^{T^{1,0}Z} = P^{TZ} \nabla^{T^{1,0}M}$. Next, we prove

that T is a $(1, 1)$ -form. Since T is real, it needs only to prove that T vanish on $T^{1,0}M \times T^{1,0}M$. By conjugation, T also vanish on $T^{0,1}M \times T^{0,1}M$.

Under the isomorphism

$$T^{1,0}M = T^{H,(1,0)}M \oplus T^{1,0}Z \simeq \pi^*T^{1,0}B \oplus T^{1,0}Z,$$

let us do some calculations. In the following, $\langle \cdot, \cdot \rangle_{g^{TZ}}$ denotes the Hermitian metric induced by g^{TZ} , same notation for $\langle \cdot, \cdot \rangle_{g^{TM}}$.

For any $V_1, V_2 \in C^\infty(M, T^{1,0}B)$, $Y \in C^\infty(M, TZ)$,

$$\begin{aligned} \langle T(V_1^H, V_2^H), Y \rangle_{g^{TZ}} &= -\langle P^{TZ}[V_1^H, V_2^H], Y \rangle_{g^{TZ}} = -\langle [V_1^H, V_2^H], Y \rangle_{g^{TM}} \\ &\stackrel{\substack{\nabla^{TM} \\ \text{no torsion}}}{=} -\langle \nabla_{V_1^H}^{TM} V_2^H - \nabla_{V_2^H}^{TM} V_1^H, Y \rangle_{g^{TM}} \\ &= -\langle \nabla_{V_1^H}^{TM} V_2^H, Y \rangle_{g^{TM}} + \langle \nabla_{V_2^H}^{TM} V_1^H, Y \rangle_{g^{TM}} \\ &\stackrel{\substack{\nabla^{TM} \\ \text{Hermitian}}}{=} -\left[V_1^H \langle V_2^H, Y \rangle_{g^{TM}} - \langle V_2^H, \nabla_{V_1^H}^{TM} Y \rangle_{g^{TM}} \right] \\ &\quad + \left[V_2^H \langle V_1^H, Y \rangle_{g^{TM}} - \langle V_1^H, \nabla_{V_2^H}^{TM} Y \rangle_{g^{TM}} \right] \quad (*) \end{aligned}$$

As the horizontal lift is compatible with conjugate, $V_1^H, V_2^H \in C^\infty(M, T^{H,(0,1)}M)$, as $T^H M \perp TZ$ w.r.t. the Hermitian metric $\langle \cdot, \cdot \rangle_{g^{TM}}$, we get

$$\begin{aligned} (*) &= \langle V_2^H, \nabla_{V_1^H}^{TM} Y \rangle_{g^{TM}} - \langle V_1^H, \nabla_{V_2^H}^{TM} Y \rangle_{g^{TM}} \\ &= \langle V_2^H, \nabla_Y^{TM} \overline{V_1^H} - [\overline{V_1^H}, Y] \rangle_{g^{TM}} - \langle V_1^H, \nabla_Y^{TM} \overline{V_2^H} - [\overline{V_2^H}, Y] \rangle_{g^{TM}} \\ &= \langle V_2^H, \nabla_Y^{TM} \overline{V_1^H} \rangle_{g^{TM}} - \langle V_1^H, \nabla_Y^{TM} \overline{V_2^H} \rangle_{g^{TM}} = 0 \end{aligned}$$

For any $V_1 \in C^\infty(M, T^{1,0}B)$, $X \in C^\infty(M, T^{1,0}Z)$, $Y \in C^\infty(M, TZ)$,

$$\begin{aligned} \langle T(V_1^H, X), Y \rangle_{g^{TZ}} &= \langle \nabla_{V_1^H}^{TZ} X - [V_1^H, X], Y \rangle_{g^{TZ}} = \langle P^{TZ} \nabla_{V_1^H}^{TM} X - [V_1^H, X], Y \rangle_{g^{TZ}} \\ &= \langle P^{TZ} (\nabla_{V_1^H}^{TM} X - [V_1^H, X]), Y \rangle_{g^{TZ}} = \langle P^{TZ} \nabla_X^{TM} V_1^H X, Y \rangle_{g^{TZ}} \\ &= \langle \nabla_X^{TM} V_1^H X, Y \rangle_{g^{TM}} = -\langle V_1^H, \nabla_{\overline{X}}^{TM} Y \rangle_{g^{TM}} \\ &\stackrel{[X,Y] \in TZ}{=} -\langle \underbrace{V_1^H}_{\in T^{1,0}M}, \underbrace{\nabla_Y^{TM} \overline{X}}_{\in T^{0,1}M} \rangle_{g^{TM}} = 0. \end{aligned}$$

□

Set $\omega^Z = \omega|_{TZ \times TZ}$, $\omega^H = \omega|_{T^H M \times T^H M}$. Extend ω^Z, ω^H as zero for other parts of $TM \times TM$. Thus, it is easy to see

$$\omega = \omega^Z + \omega^H.$$

Lemma. For $V \in C^\infty(B, TB)$, the Lie derivative $\mathcal{L}_{V^H}^Z \omega^Z = 0$ where V^H is the horizontal lift of V .

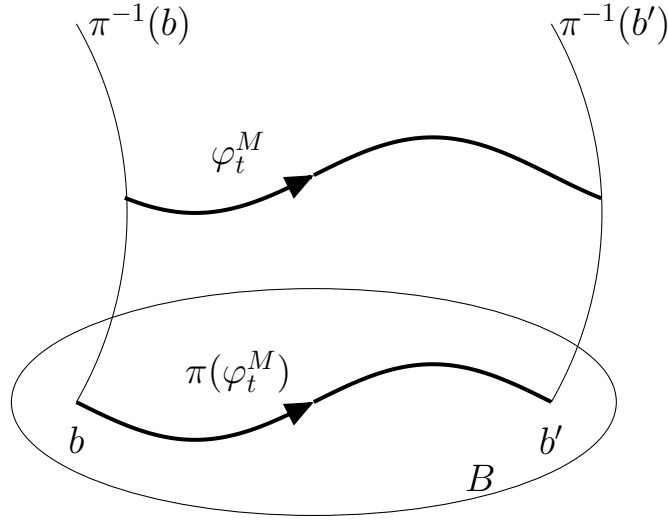
Proof. The supernote Z on $\mathcal{L}$ means that the flot φ_t^M of V^H sends fiber to fiber. To see this, for any $x \in M$, φ_t^M satisfies the system

$$\begin{cases} \frac{\partial}{\partial t} \varphi_t^M(x) &= V^H(\varphi_t^M(x)) \\ \varphi_0^M(x) &= x \end{cases}$$

where $t \in (-\epsilon, \epsilon)$ for some $\epsilon > 0$. Apply π to the system, we obtain

$$\begin{cases} \frac{\partial}{\partial t} \pi(\varphi_t^M)(\pi(x)) &= V(\pi(\varphi_t^M)(\pi(x))) \\ \pi(\varphi_0^M)(\pi(x)) &= \pi(x) \end{cases}$$

Thus, if we denote by φ_t the flot of V , then $\pi\varphi_t^M = \varphi_t\pi$. The flot φ_t^M of V^H sends fiber to fiber. See the figure below:



Therefore, $\mathcal{L}_{V^H}^Z \omega^Z = \frac{\partial}{\partial t}|_{t=0} (\varphi_t^M)^* \omega^Z$ is well-defined. By Cartan's formula,

$$\mathcal{L}_{V^H}^Z \omega^Z = i_{V^H} d\omega^Z + di_{V^H} \omega^Z = i_{V^H} d\omega^Z \stackrel{d\omega=0}{=} -i_{V^H} d\omega^H$$

And for any $X, Y \in C^\infty(M, TZ)$,

$$\begin{aligned} i_{V^H} d\omega^H(X, Y) &= (\mathcal{L}_{V^H} \omega^H - di_{V^H} \omega^H)(X, Y) \\ &= V^H(\omega^H(X, Y)) - \omega^H([V^H, X], Y) - \omega^H(X, [V^H, Y]) \\ &\quad - [V^H(\omega^H(X, Y)) - X(\omega^H(V^H, Y)) + Y(\omega^H(V^H, X))] \\ &= 0 \end{aligned}$$

□

The geometric preparation for this chapter is complete.

§3.2 Holomorphic Hermitian Connection on infinite dimensional bundle

Let $\pi : M \rightarrow B$ be a holomorphic submersion of complex manifold with compact fiber Z . From now on, we assume that $(\pi, g^{TZ}, T^H M)$ is Kahler fibration and $\omega = \omega^Z + \omega^H$ is an associated real closed $(1, 1)$ -form. Let (ξ, h^ξ) be a holomorphic vector bundle on M with Hermitian metric h^ξ . Denote by ∇^ξ be Chern connection on (ξ, h^ξ) and $R^\xi = \nabla^{\xi, 2} \in \Omega^{1, 1}(M, \text{End}(\xi))$. Extend $\mathbb{C}$ -linearly the Riemannian metric g^{TZ} , denoted also by g^{TZ} . Thus, it is a $(1, 1)$ -form and defines an identification

$$\begin{aligned} T^{1,0}Z &\xrightarrow{\cong} T^{*(0,1)}Z \\ U &\mapsto g^{TZ}(U, \cdot) \xlongequal{\text{denoted by}} U^* \end{aligned}$$

The dual anti-holomorphic bundle $T^{(0,1)*}Z$ inherits the holomorphic structure of $T^{1,0}Z$ and the Chern connection $\nabla^{T^{1,0}Z}$ induces the corresponding holomorphic Hermitian (Chern) connection $\nabla^{T^{(0,1)*}Z}$ on $T^{(0,1)*}Z$. Precisely, for $X \in C^\infty(Z, TZ)$, $U^* \in C^\infty(Z, T^{(0,1)*}Z)$, $\nabla_X^{T^{(0,1)*}Z} U^* = \left(\nabla_X^{T^{1,0}Z} U \right)^*$.

Let $\nabla^{\Lambda(T^{(0,1)*}Z) \otimes \xi}$ be the Chern connection on $\Lambda(T^{(0,1)*}Z) \otimes \xi$ induced by $\nabla^{T^{(0,1)*}Z}$ and ∇^ξ .

For any $y \in B$, let $Z_y = \pi^{-1}(y)$ and

$$E_y^p = \Omega^{0,p}(Z_y, \xi|_{Z_y}) = C^\infty(Z_y, \Lambda T^{(0,1)*}Z_y \otimes \xi|_{Z_y}).$$

Set $l = \dim_{\mathbb{C}} Z$. Precisely,

$$E_y = \bigoplus_{p=0}^l E_y^p.$$

Also, set $E_y^+ = \bigoplus_{p \text{ even}} E_y^p$, $E_y^- = \bigoplus_{p \text{ odd}} E_y^p$.

We use in formal way $\{E_y\}_{y \in B}$ as an infinite dimensional vector bundle on B . Set

$$F_p = \Lambda^p(T^{(0,1)*}Z) \otimes \xi, \quad F = \bigoplus_{p=0}^l F_p.$$

Certainly,

$$C^\infty(B, E^p) = C^\infty(M, F^p), \quad C^\infty(B, E) = C^\infty(M, F).$$

and for each $y \in B$, we have a chain complex $(E_y, \bar{\partial}^Z)$ as follows:

$$0 \rightarrow E_y^0 \xrightarrow{\bar{\partial}^Z} E_y^1 \xrightarrow{\bar{\partial}^Z} \cdots \xrightarrow{\bar{\partial}^Z} E_y^l \rightarrow 0$$

where $\bar{\partial}^Z$ is the Dolbeault operator w.r.t. Z part.
Let dv_Z be the Riemannian volume form of (Z, g^{TZ}) , then

$$dv_Z = \frac{(\omega^Z)^l}{l!} = \frac{1}{l!} \underbrace{\omega^Z \wedge \omega^Z \wedge \cdots \wedge \omega^Z}_{l \text{ times}}.$$

For each $y \in B$, define an L^2 -product on E_y by

$$\langle s, s' \rangle_{E_y} = \int_{x \in Z_y} \langle s(x), s'(x) \rangle_{\Lambda T^{(0,1)*} Z_y \otimes \xi|_{Z_y}} dv_{Z_y}(x)$$

where $s, s' \in E_y$, $dv_{Z_y} = \frac{1}{l!} (\omega^Z|_{Z_y})^l$. Thus, for $s_1, s_2 \in C^\infty(B, E)$, define a function $\langle s_1, s_2 \rangle_E$ on B by

$$\langle s_1, s_2 \rangle_E(y) = \langle s_1(y), s_2(y) \rangle_{E_y}.$$

The fiberwise Dolbeault operator $\bar{\partial}^Z$ has a formal adjoint $(\bar{\partial}^Z)^*$ w.r.t. $\langle \cdot, \cdot \rangle_{E_y}$ for each $y \in B$. Define fiberwisely

$$D = \bar{\partial}^Z + (\bar{\partial}^Z)^*, \quad D_\pm = D|_{E_\pm}$$

At each point $y \in B$, $D_y = \begin{bmatrix} 0 & D_- \\ D_+ & 0 \end{bmatrix}$ is self-adjoint on E_y . Recall the Clifford action $c(\cdot)$ on $F = \Lambda(T^{(0,1)*} Z) \otimes \xi$. For $u \in T^{1,0} Z$ with identification $u^* \in T^{(0,1)*} Z$,

$$c(u) = \sqrt{2}u^* \wedge, \quad c(\bar{u}) = -\sqrt{2}i_{\bar{u}} \in \text{End}\left(\Lambda(T^{(0,1)*} Z) \otimes \xi\right)$$

Proposition. $\sqrt{2}D$ is the Dirac operator, i.e.

$$\sqrt{2}D = \sum_{j=1}^{2l} c(e_j) \nabla_{e_j}^F$$

where e_j is an orthonormal frame of (TZ, g^{TZ}) .

Proof. As ∇^F is holomorphic, $\bar{\partial}^Z = \sum_{j=1}^l \bar{\omega}^j \wedge \nabla_{\bar{\omega}^j}^F$. We calculate the adjoint operator. For any s_1, s_2 with $s_1 \in \Lambda^{\bullet+1}(T^{(0,1)*} Z) \otimes \xi$ and $s_2 \in \Lambda^\bullet(T^{(0,1)*} Z) \otimes \xi$, we calculate (omit $\sum_{j=1}^l$)

$$\begin{aligned} \langle (\bar{\partial}^Z)^* s_1, s_2 \rangle_F &= \langle s_1, \bar{\partial}^Z s_2 \rangle_F = \langle s_1, \bar{\omega}^j \wedge \nabla_{\bar{\omega}^j}^F \rangle_F \\ &= \langle i_{\bar{\omega}^j} s_1, \nabla_{\bar{\omega}^j}^F s_2 \rangle_F \\ &= \omega_j \langle i_{\bar{\omega}^j} s_1, s_2 \rangle_F - \langle \nabla_{\omega_j}^F i_{\bar{\omega}^j} s_1, s_2 \rangle_F \\ &= -\langle i_{\bar{\omega}^j} \nabla_{\omega_j}^F s_1, s_2 \rangle_F - \langle [\nabla_{\omega_j}^F, i_{\bar{\omega}^j}] s_1, s_2 \rangle_F + \omega_j \langle i_{\bar{\omega}^j} s_1, s_2 \rangle_F \quad (*) \end{aligned}$$

Recall the relations

$$\begin{aligned} e_{2j-1} &= \frac{1}{\sqrt{2}}(\omega_j + \bar{\omega}_j) \\ e_{2j} &= \frac{\sqrt{-1}}{\sqrt{2}}(\omega_j - \bar{\omega}_j) \\ \omega_j &= \frac{1}{\sqrt{2}}(e_{2j-1} - \sqrt{-1}e_{2j}) \\ \bar{\omega}_j &= \frac{1}{\sqrt{2}}(e_{2j-1} + \sqrt{-1}e_{2j}) \end{aligned}$$

On the book Proposition 1.3.1, in the equation $[\nabla_V^F, c(W)] = c(\nabla_V^{TZ}W)$, take $V = e_{2j-1}, e_{2j}$ respectively, take $W = e_{2j-1}$ and consider only the part sending degree $(\bullet, \bullet)$ to $(\bullet, \bullet - 1)$,

$$[\nabla_{e_{2j-1}}^F, i_{\bar{\omega}_j}] = i_{\nabla_{e_{2j-1}}^{T^{0,1}Z}\bar{\omega}_j}, \quad [\nabla_{e_{2j}}^F, i_{\bar{\omega}_j}] = i_{\nabla_{e_{2j}}^{T^{0,1}Z}\bar{\omega}_j}$$

So, $[\nabla_{\omega_j}^F, i_{\bar{\omega}_j}] = i_{\nabla_{\omega_j}^{T^{0,1}Z}\bar{\omega}_j}$. Consider the 1-form $\alpha(\cdot) = \langle c(\cdot)s_1, s_2 \rangle_{F|_x}$ on Z where $x \in Z$.

$$\begin{aligned} \text{Tr}(\nabla^F \alpha) &= \left[e_{2j-1}(\alpha(e_{2j-1})) - \alpha(\nabla_{e_{2j-1}}^{TZ} e_{2j-1}) \right] + \left[e_{2j}(\alpha(e_{2j})) - \alpha(\nabla_{e_{2j}}^{TZ} e_{2j}) \right] \\ &= \left[e_{2j-1} \langle -i_{\bar{\omega}_j} s_1, s_2 \rangle_{F|_x} - \langle c(\nabla_{e_{2j-1}}^{TZ} e_{2j-1}) s_1, s_2 \rangle_{F|_x} \right] + \left[e_{2j} \langle \sqrt{-1} i_{\bar{\omega}_j} s_1, s_2 \rangle_{F|_x} - \langle c(\nabla_{e_{2j}}^{TZ} e_{2j}) s_1, s_2 \rangle_{F|_x} \right] \\ &= (-e_{2j-1} + \sqrt{-1}e_{2j}) \langle i_{\bar{\omega}_j} s_1, s_2 \rangle_{F|_x} - \langle -i_{\nabla_{e_{2j-1}}^{T^{0,1}Z}\bar{\omega}_j} s_1, s_2 \rangle_{F|_x} - \langle -i_{\nabla_{e_{2j}}^{T^{0,1}Z}(-\sqrt{-1})\bar{\omega}_j} s_1, s_2 \rangle_{F|_x} \\ &= -\sqrt{2}\omega_j \langle i_{\bar{\omega}_j} s_1, s_2 \rangle_{F|_x} + \sqrt{2} \langle i_{\nabla_{\omega_j}^{T^{0,1}Z}\bar{\omega}_j} s_1, s_2 \rangle_{F|_x} \end{aligned}$$

The integral of $\text{Tr}(\nabla \alpha)$ on Z is zero. So,

$$(*) = -\langle i_{\bar{\omega}_j} \nabla_{\omega_j}^F s_1, s_2 \rangle_F = \langle -i_{\bar{\omega}_j} \nabla_{\omega_j}^F s_1, s_2 \rangle_F.$$

Therefore,

$$(\bar{\partial}^Z)^* = \sum_{j=1}^l -i_{\bar{\omega}_j} \nabla_{\omega_j}^F$$

CONFUSION

I AM CONFUSED. WHY I DID NOT USE THE KAHLER CONDITION OF ω^Z BUT I OBTAIN THE EQUATION ABOVE.

It is easy to check the rest. □

The following content is based on

Professor Ma Xiaonan's instruction on October 14th 2025

Let F be the holomorphic vector bundle on M defined by

$$F|_x = \Lambda T_x^{(0,1)*} Z_{\pi(x)} \otimes \xi|_x$$

Definition. Define $C^\infty(B, E) := C^\infty(M, F)$. Define the connection $\tilde{\nabla}$ on E :

$$\begin{aligned} \tilde{\nabla} : C^\infty(TB) \times C^\infty(B, E) &\rightarrow C^\infty(B, E) \\ (U, s) &\mapsto \nabla_{U^H}^F s \end{aligned}$$

If $s_1, s_2 \in C^\infty(B, E)$, the Hermitian metric $\langle s_1, s_2 \rangle_E$ at $y \in B$ is defined by

$$\langle s_1, s_2 \rangle_E(y) = \int_{x \in Z_y} \langle s_1(x), s_2(x) \rangle_F dv_Z(x)$$

where $dv_Z = \frac{1}{l!}(\omega^Z)^l$ is the volume form restricted on $Z_y = \pi^{-1}(y)$.

Proposition. *The connection $\tilde{\nabla}$ is Hermitian for $\langle \cdot, \cdot \rangle_E$. Furthermore,*

$$\tilde{\nabla}^2(V_1, V_2) = R^F(V_1^H, V_2^H) - \nabla_{T(V_1^H, V_2^H)}^F.$$

Proof. For $y \in B$, $s_1, s_2 \in C^\infty(B, E)$,

$$\begin{aligned} \langle \tilde{\nabla}_U s_1, s_2 \rangle_E(y) &= \int_{x \in Z_y} \langle \nabla_{U^H}^F s_1, s_2 \rangle_F(x) dv_Z(x) \\ &= \int_{Z_y} \left[\mathcal{L}_{U^H} \left(\langle s_1, s_2 \rangle_F dv_Z \right) \right] - \int_{Z_y} \langle s_1, s_2 \rangle_F \underbrace{\mathcal{L}_{U^H} dv_Z}_{=0} \\ &\quad - \int_{Z_y} \langle s_1, \nabla_{U^H}^F s_2 \rangle_F dv_Z \\ &= \frac{\partial}{\partial t} \Big|_{t=0} \int_{Z_y} \left[(e^{tU^H})^* \left(\langle s_1, s_2 \rangle_F dv_Z \right) \right] - \langle s_1, \tilde{\nabla}_U s_2 \rangle_E \\ &\stackrel{\text{flot } e^{U^H}}{\text{fiber to fiber}} = \frac{\partial}{\partial t} \Big|_{t=0} (e^{tU})^* \int_{Z_{e^t U y}} \langle s_1, s_2 \rangle_F dv_Z \\ &= U \langle s_1, s_2 \rangle_E. \end{aligned}$$

For $V_1, V_2 \in C^\infty(B, TB)$, as the flot $e^{tV_1^H}$ preserves fibers, $\pi_*[V_1^H, V_2^H] = [V_1, V_2]$. Thus,

$$[V_1^H, V_2^H] = P^{TZ}[V_1^H, V_2^H] + [V_1, V_2]^H.$$

For $s \in C^\infty(B, E)$,

$$\begin{aligned} \tilde{\nabla}^2(V_1, V_2)s &= \tilde{\nabla}_{V_1} \tilde{\nabla}_{V_2} s - \tilde{\nabla}_{V_2} \tilde{\nabla}_{V_1} s - \tilde{\nabla}_{[V_1, V_2]} s \\ &= \nabla_{V_1^H}^F \nabla_{V_2^H}^F s - \nabla_{V_2^H}^F \nabla_{V_1^H}^F s - \nabla_{[V_1, V_2]^H}^F s \\ &= R^F(V_1^H, V_2^H)s + \nabla_{P^{TZ}[V_1^H, V_2^H]}^F s \\ &= R^F(V_1^H, V_2^H)s - \nabla_{T(V_1^H, V_2^H)}^F s \end{aligned}$$

Since $\tilde{\nabla}$ is Hermitian, its curvature form $\tilde{\nabla}^2(\cdot, \cdot)$ is automatically skew-adjoint. Also as the connection ∇^F is the Chern connection, its curvature form is $(1, 1)$ -type. Besides, $T((\cdot)^H, (\cdot)^H)$ is $(1, 1)$ -type. Therefore, $\tilde{\nabla}^2$ is $(1, 1)$ -type. Let $m = \dim_{\mathbb{C}} B$, $(f_1, f_2, \dots, f_{2m})$ be a frame of TB , $\{g_\alpha = \frac{1}{\sqrt{2}}(f_{2\alpha-1} - \sqrt{-1}f_{2\alpha})\}_{1 \leq \alpha \leq m}$ be the corresponding frame of $T^{1,0}B$.

$$\sqrt{2}D = \sqrt{2}(\bar{\partial}^Z + \bar{\partial}^{Z,*}) \stackrel{\omega^Z \text{ is Kahler}}{=} \sum_{l=1}^{2l} c(e_j) \nabla_{e_j}^F$$

where $\{e_j\}_{1 \leq j \leq 2l}$ is an orthogonal frame of TZ .

CONFUSION

SAME CONFUSION AS ABOVE. WHERE HAVE I USED THE KÄHLER CONDITION?

Let us calculate (omit $\sum_{\alpha=1}^{2m}, \sum_{j=1}^{2l}, \sum_{k=1}^{2l}$)

$$\begin{aligned}
[\tilde{\nabla}, \sqrt{2}D] &= f^\alpha \wedge \nabla_{f_\alpha^H}^F c(e_j) \nabla_{e_j}^F + c(e_j) \nabla_{e_j}^F f^\alpha \wedge \nabla_{f_\alpha^H}^F \\
\underline{\underline{\nabla_{e_j}^F f^\alpha \wedge f^\alpha \wedge \nabla_{e_j}^F}} &= f^\alpha \wedge \nabla_{f_\alpha^H}^F c(e_j) \nabla_{e_j}^F + c(e_j) f^\alpha \wedge \nabla_{e_j}^F \nabla_{f_\alpha^H}^F \\
&= f^\alpha \wedge [\nabla_{f_\alpha^H}^F, c(e_j)] \nabla_{e_j}^F + f^\alpha \wedge c(e_j) \nabla_{f_\alpha^H}^F \nabla_{e_j}^F + c(e_j) f^\alpha \wedge \nabla_{e_j}^F \nabla_{f_\alpha^H}^F \\
\underline{\underline{[\nabla_{f_\alpha^H}^F, c(e_j)] = c(\nabla_{f_\alpha^H}^{TZ} e_j)}} &= f^\alpha \wedge c(\nabla_{f_\alpha^H}^{TZ} e_j) \nabla_{e_j}^F + f^\alpha \wedge c(e_j) [\nabla_{f_\alpha^H}^F \nabla_{e_j}^F - \nabla_{e_j}^F \nabla_{f_\alpha^H}^F] \\
&= [f^\alpha \wedge c(e_k) \langle \nabla_{f_\alpha^H}^{TZ} e_j, e_k \rangle_{g^{TZ}} \nabla_{e_j}^F] + f^\alpha \wedge c(e_j) [\nabla_{f_\alpha^H}^F \nabla_{e_j}^F - \nabla_{e_j}^F \nabla_{f_\alpha^H}^F] \\
&= [-f^\alpha \wedge c(e_k) \langle e_j, \nabla_{f_\alpha^H}^{TZ} e_k \rangle_{g^{TZ}} \nabla_{e_j}^F] + f^\alpha \wedge c(e_j) [\nabla_{f_\alpha^H}^F \nabla_{e_j}^F - \nabla_{e_j}^F \nabla_{f_\alpha^H}^F] \\
&= [-f^\alpha \wedge c(e_k) \nabla_{\nabla_{f_\alpha^H}^{TZ} e_k}^F] + f^\alpha \wedge c(e_j) [\nabla_{f_\alpha^H}^F \nabla_{e_j}^F - \nabla_{e_j}^F \nabla_{f_\alpha^H}^F] \\
&= f^\alpha \wedge c(e_j) [\nabla_{f_\alpha^H}^F \nabla_{e_j}^F - \nabla_{e_j}^F \nabla_{f_\alpha^H}^F] - f^\alpha \wedge c(e_j) \nabla_{\nabla_{f_\alpha^H}^{TZ} e_j}^F \\
&= f^\alpha \wedge c(e_j) \left[R^F(f_\alpha^H, e_j) - \nabla_{\nabla_{f_\alpha^H}^{TZ} e_j - [f_\alpha^H, e_j]}^F \right] \\
&= f^\alpha \wedge c(e_j) \left[R^F(f_\alpha^H, e_j) - \nabla_{T(f_\alpha^H, e_j)}^F \right] \quad (*)
\end{aligned}$$

We know the identity

$$c(e_{2j-1})\varphi(e_{2j-1}) + c(e_{2j})\varphi(e_{2j}) = c(\omega_j)\varphi(\overline{\omega_j}) + c(\overline{\omega_j})\varphi(\omega_j)$$

holds for every $1 \leq j \leq l$ and every linear map φ . A similar identity holds for $f_{2\alpha-1}, f_{2\alpha}$ and $g_\alpha, \overline{g}_\alpha$. Thus,

$$(*) = \sum_{\alpha=1}^m \sum_{j=1}^l g^\alpha \wedge c(w_j) \left[R^F(g_\alpha^H, \overline{w_j}) - \nabla_{T(g_\alpha^H, \overline{w_j})}^F \right] + \overline{g}^\alpha \wedge c(\overline{w_j}) \left[R^F(\overline{g}_\alpha^H, w_j) - \nabla_{T(\overline{g}_\alpha^H, w_j)}^F \right] \quad (**)$$

On the other side, $\tilde{\nabla} \cdot D = \tilde{\nabla} \cdot \bar{\partial}^Z + \tilde{\nabla} \cdot \bar{\partial}^{Z,*}$. As $\tilde{\nabla} \cdot$ derive only along the direction of Z , it preserves the $\mathbb{Z}$ -grading on E , so $\tilde{\nabla} \cdot \bar{\partial}^Z$ increases the degree on E by 1, $\tilde{\nabla} \cdot \bar{\partial}^{Z,*}$ decrease the degree on E by 1. Compare with the equation (**), we get

$$\tilde{\nabla} \cdot \bar{\partial}^Z = \sum_{\alpha=1}^m \sum_{j=1}^l g^\alpha \wedge c(w_j) \left[R^F(g_\alpha^H, \overline{w_j}) - \nabla_{T(g_\alpha^H, \overline{w_j})}^F \right]$$

$$\tilde{\nabla} \cdot \bar{\partial}^{Z,*} = \overline{g}^\alpha \wedge c(\overline{w_j}) \left[R^F(\overline{g}_\alpha^H, w_j) - \nabla_{T(\overline{g}_\alpha^H, w_j)}^F \right]$$

This implies that $\forall U \in C^\infty(B, T^{1,0}B)$, $\tilde{\nabla}_{\overline{U}} \cdot \bar{\partial}^Z$. □

Remark. If (F, h^F, ∇^F) is a finite dimensional Hermitian vector bundle on a complex manifold B with Hermitian connection ∇^F and its curvature $R^F = \nabla^{F,2} \in \Omega^{1,1}(B, \text{End}F)$, then we have the Koszul-Malgrange Theorem:

There exists a unique holomorphic structure on F such that ∇^F is the Chern connection on (F, h^F) , i.e.

$$\forall s \in C^\infty(B, F), \quad \bar{\partial}^F s := (\nabla^F s)^{0,1}.$$

However, in our setting, the vector bundle E over B has an infinite dimension, $\tilde{\nabla}$ is a Hermitian connection and $\tilde{\nabla}^2 \in \Omega^{1,1}(B, \text{End}(E))$. Formally, E is a holomorphic vector bundle, $\bar{\partial}^Z$ can be seen as a “holomorphic” section of “holomorphic” vector bundle $\text{End}(E)$. Let us prove the Koszul-Malgrange Theorem for line bundle.

If F is a line bundle, we define $\bar{\partial}^F s := (\nabla^F)^{0,1} s$. Let s_0 be a local frame of F in an open neighbourhood U of x_0 . Let $a \in \Omega^{0,1}(U)$ given by $(\nabla^F)^{0,1} s_0 = a \otimes s_0$.

$$\begin{aligned} 0 &= \bar{\partial}^{F,2} s = ((\nabla^F)^{0,1})^2 s = (\nabla^F)^{0,1} (a \otimes s) \\ &= (\bar{\partial} a) s_0 - a (\nabla^F)^{0,1} s = (\bar{\partial} a) s_0 - a \wedge a \otimes s \\ &= (\bar{\partial} a) s_0. \end{aligned}$$

Thus, $\bar{\partial} a = 0$ on U . There exists an open subset U_0 of U and $b \in C^\infty(U_0)$ such that $a = \bar{\partial} b$,

$$\begin{aligned} \bar{\partial}^F (e^{-b} s_0) &= -e^{-b} \bar{\partial} b \otimes s_0 + e^{-b} (\nabla^F)^{0,1} s \\ &= e^{-b} [-a \otimes s_0 + (\nabla^F)^{0,1} s] = 0. \end{aligned}$$

So, $e^{-b} s_0$ defines a holomorphic frame of $F|_{U_0}$.

CONFUSION

I AM CONFUSED about the existence of b . Is it because U is a simple connected domain (or product of discs) which implies the degeneration of $(0, 1)$ -part of Dolbeault cohomology?

§3.3 C^∞ determinant line bundle

We keep the setting above. $\sqrt{2}D \xrightarrow{\omega^Z \text{ is Kahler}} = \sum_{j=1}^{2l} c(e_j) \nabla_{e_j}^F$.

$$2D^2 = \Delta^F + \underbrace{R^\xi(w_j, \bar{w}_j) + \frac{1}{2}(R^\xi + \frac{1}{2}\text{Tr}[R^{T^{1,0}Z}])(w_i, \bar{w}_j)\bar{w}^j \wedge i_{\bar{w}_i}}_{\text{noted by } Q, \text{ zero order part}}$$

where $\Delta^F = -\sum_j (\nabla_{e_j}^F)^2 - \nabla_{\nabla_{e_j}^{TZ} e_j}^F$ Laplacian associated to ∇^F . $2D^2$ is a generalized Laplacian on F , it is a second order elliptic differential operator.

CONFUSION

I TRIED TO APPLY THE EQUATION 1.3.26 OF PROFESSOR'S BOOK, BUT I CANNOT OBTAIN THE HALF AND THE QUATER COEFFICIENTS.

Theorem. *For the compactness of Z , the spectrum $\text{Spec}(D^2)$ of D^2 is discrete and there exists an orthonormal basis $\{\varphi_j\}_{j=1}^\infty$ of $L^2(Z, F)$ such that*

$$D^2\varphi_j = \lambda_j\varphi_j, \quad 0 \leq \lambda_1 \leq \lambda_2 \leq \dots \rightarrow +\infty.$$

And $\dim \text{Ker}(D^2|_{\Omega^{0,k}} - \lambda_j) < +\infty$, $\varphi_j \in C^\infty(Z, F)$.

Let $E_{\mu,y}^p$ the eigenspace of $D_{y|\Omega^{0,p}(Z_y, \xi|_{Z_y})}^2$ associated to the eigenvalue μ . For $b > 0$, let $K_y^{b,p} = \bigoplus_{\mu < b} E_{\mu,y}^p$. By the theorem above, $K_y^{b,p}$ is a finite dimensional subspace of $E_y^p = \Omega^{0,p}(Z_y, \xi_{Z_y})$. Set

$$U^b = \{y \in B \mid b \notin \text{Spec}(D_y^2)\}.$$

CONFUSION

I JUST COPIED THIS PART TO FAST IN CLASS, I CAN-
NOT FIGURE IT OUT.

Theorem. *U^b is open, $K^{b,p} \bigcup_{y \in U^b} K_y^{b,p}$ is a smooth finite dimensional vector bundle on U^b .*

Proof. For $y \in U^b$, let $P_y^b :$

□